



A Doppler frequency metric for GNSS spoofing detection in urban environments

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Abstract

A reliable navigation solution is increasingly critical as Global Navigation Satellite System (GNSS) technology underpins applications such as transportation. However, spoofing attacks remain a significant threat by injecting misleading position, velocity, and time (PVT) estimates into GNSS receivers. Although various detection techniques have been proposed, many rely on hardware modifications, additional sensors, or computationally intensive algorithms, which may limit their practicality for low-cost devices. This gap highlights the need for lightweight, receiver-based detection methods that exploit existing measurements without requiring additional resources. This paper proposes a spoofing detection approach that operates solely on standard receiver outputs, specifically Doppler frequency, and monitors the Doppler rate feature under attacks. The core principle is to compare the measured Doppler frequency shift rate from receiver tracking outputs with the predicted Doppler frequency shift rate derived from receiver navigation results, where a statistically significant inconsistency indicates a potential spoofing attack. We analyze detection performance under various user dynamics and spoofer antenna configurations, covering 16 representative attack scenarios, and demonstrate effectiveness in 11 out of them. To distinguish spoofing from noise and unintentional interference, we design an adaptive threshold and adopt the Neyman-Pearson decision rule to maximise the probability of detection for a prescribed false-alarm rate. The method is validated in simulation and on public spoofing-attack datasets. In the TEXBAT validation, the proposed method achieves a detection delay of less than 5 s while maintaining a false alarm rate of approximately 2%, demonstrating its reliability under representative spoofing attacks.

Keywords Global Navigation Satellite System (GNSS) · Spoofing detection · Doppler frequency · Signal processing

1 Introduction

Global Navigation Satellite System (GNSS) signals are vulnerable to spoofing because of their low received power and publicly known signal and modulation formats [1]. Attackers can generate, relay, or selectively delay GNSS-like signals to mislead a receiver's position, velocity, and time (PVT) solution [2]. These attacks can be mounted with inexpensive programmable transmitters and commercial-off-the-shelf (COTS) hardware [3, 4], raising security concerns for mass-market applications (e.g., smartphones and vehi-

cles) where reliable positioning underpins location-based services (LBS).

Numerous spoofing detection methods incorporate external hardware assistance. Inertial navigation systems (INS) that are resilient to GNSS spoofing interference have been developed in various integration schemes with GNSS. Detection of spoofing attacks can be achieved by checking inconsistencies between GNSS spoofed measurements and INS measurements, such as acceleration, movement direction, pseudorange measurement, or position [5–7]. Furthermore, the utilization of multi-antenna receivers enables the detection or rejection of spoofing signals through monitoring angle-of-arrival [8, 9] or frequency difference of arrival [10] of the incoming signals. However, this paper targets stand-alone, low-cost receivers and focuses on intermediate spoofing that avoids conspicuous power surges or obvious tracking anomalies during injection and takeover. In general, there are two categories of spoofing defense techniques on

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the receiver's side: 1) the baseband processing algorithms, and 2) the detection methods using receiver measurements.

The baseband-level detection methods typically analyze receiver correlator outputs. Under synchronous spoofing, authentic and spoofed signals are received and processed by the receiver simultaneously, producing deviations from the nominal correlation function shape. The signal quality monitoring (SQM) tests effectively detect such peak distortions [11–13]. However, the correlation function may remain distortion-free in scenarios where the spoofing signal has greatly high power or significant code delay. Hence, detectors that integrate both signal power and correlation distortion features have been proposed [14, 15]. Using additional correlators broadens the detectable range and improves delay-domain resolution, motivating numerous multi-correlator methods, including correlation-shape or symmetry monitoring [16–18], maximum-likelihood (ML) estimation of spoofer parameters [19, 20], and image-based characterization of correlation distortion and power anomalies [21]. However, the detection performance increases at the expense of additional correlators and augmented computational load for implementation. Distortion-based tests also face practical challenges. First, false alarms may arise due to multipath effects, as the reflected signal could induce similar correlation distortion. Second, the assumption of spoofing-absence data is usually required as prior knowledge for detecting abnormalities. Last, correlation distortion-free may also exist in scenarios where only spoofing signals are tracked, or when the spoofer successfully gains control of the receiver, causing the tracking point to drift beyond the code chip length.

Measurement-level detection methods use receiver-level observables and solutions, such as the receiver PVT, pseudorange, Doppler frequency, or carrier-to-noise ratio (C/N_0). These approaches are computationally light and can be deployed on consumer-grade receivers with minimal firmware changes. For example, traditional receiver autonomous integrity monitoring (RAIM) detects spoofing signals that fail to produce a self-consistent navigation solution. However, RAIM may be ineffective against the spoofer that generates all spoofing signals based on a false state and produces self-consistent range measurements [3]. Radio power monitoring metrics, such as automatic gain control [22] and C/N_0 measurements [23] or a combination thereof [24, 25], have also been employed. However, power-change detectors are most informative during the injection phase; once takeover is complete and power levels stabilize, their sensitivity diminishes.

Doppler measurement is one of the most common parameters that assess the relative motion between the transmitter and receiver, indicating the incoming signal's spatial characteristics. For a single-antenna spoofer, identical kinematics across spoofed channels produce nearly the same Doppler bias for multiple satellites; detecting these shared biases can indicate an attack [26]. A Doppler frequency differ-

ence model is proposed, which exhibits different linearity performances in the spoofing absence versus presence cases [27]. However, such methods can fail when the adversary uses multiple transmitters or a phased array, introducing satellite-dependent Doppler biases. More recently, Zhou et al. proposed a velocity-consistency check that compares the velocity estimated from pseudorange rates with velocity derived directly from Doppler measurements [28]. For dynamic receivers, spoofer–receiver relative motion causes these two estimates to diverge, enabling detection; for static receivers or timing-oriented attacks, the inconsistency is small, reducing sensitivity. Accurate estimation of the spoofing signal's Doppler frequency is a prerequisite for reliable detection. Some advanced spoofers can exploit spoofer–receiver geometry to manipulate carrier frequencies and align the spoofed Doppler with the authentic Doppler, thereby evading tests that rely on frequency tracking error [3]. To address this type of spoofing attack, the authors' previous work proposed a parameter estimation and correction method using Doppler measurement [29]. These factors underscore the need to identify potential attack modes, assess receiver vulnerabilities, and develop robust detection and mitigation strategies, particularly for mass-market, standalone receivers.

This paper targets intermediate spoofing attack that deceives the victim receiver quietly, by avoiding raising an abnormal power rise or tracking error. This paper proposes a single metric that considers the Doppler frequency rate features for spoofing detection. We first characterize how various spoofer implementations shape Doppler behavior: many spoofers generate carrier and code in lockstep, which can create Doppler frequency inconsistencies with authentic signals and trigger existing detectors. To avoid such tracking anomalies, some attackers may manipulate carrier phase and Doppler using precise knowledge of the receiver's PVT. In either case, a bias remains in the measured Doppler and its time derivative due to spoofer–receiver relative motion or deliberate carrier manipulation. Accordingly, we detect spoofing by monitoring the mismatch between the range-predicted Doppler rate and the receiver-measured Doppler rate.

The main contributions are summarized as follows:

- 1) A new method for spoofing detection is developed by monitoring the difference between the change rate of the receiver-measured Doppler and the predicted Doppler rate derived from a range-measurement-based solution.
- 2) The proposed detection method demonstrates efficacy across a wide range of typical spoofing attack scenarios. The hypothesis test model is constructed based on features of the proposed metric's bias and noise variance.

- 3) We build a realistic spoofing-attack simulator and validate the detector using both simulation studies and real data from public datasets.

The remainder of the paper is organized as follows. Section 2 reviews spoofing attack modes. Section 3 presents the proposed detection method and analyzes its theoretical effectiveness across different attack modes. Section 4 details the detection algorithm. Section 5 describes the spoofing-attack simulator, examines factors that influence detection performance, and reports results on real-world datasets. Finally, Section 6 concludes with remarks and future work.

2 Spoofing attack mode

Open signal modulation and structure enable spoofers to synthesize self-consistent signals conditioned on a false navigation state. Overpowering spoofers that transmit high-power signals to break the receiver's tracking loops and force re-acquisition are readily detected via abnormal power metrics. This paper focuses on more advanced synchronous spoofers that avoid obvious fluctuations in C/N_0 or loss of lock by generating spoofed spreading codes and data streams that remain synchronized with the authentic line-of-sight (LOS) signals [3]. At the start of such attacks, both authentic and spoofed signals are present at the victim antenna. To gain control of the tracking loops, the spoofer gradually increases its power advantage so that the receiver transitions to the spoofed signals. For example, Fig. 1 shows the time history of the correlator outputs in the code-delay domain under a synchronous attack: during the injection phase the spoofed code phase aligns with the authentic LOS code phase; subsequently, during the drag-off phase, the spoofer steers the receiver's code-tracking point away from the true delay, driving the navigation solution to an erroneous position. Consequently, detection during the drag-off phase is essential, because the receiver's PVT is already being compromised.

In terms of their effect on the navigation solution, spoofing attacks are commonly categorized as poisoning, timing, or a combination of both [3, 30, 31]. This paper focuses on the first two. In both cases, the spoofer reconstructs each satellite's pseudorange according to a spoofer-target false state. For example, the spoofer may inject per-channel pseudorange errors, denoted $\Delta\rho^i$, or impose a common timing bias Δt (equivalently, a constant pseudorange offset $\Delta t \cdot c$) on the receiver's measurements:

$$\rho_0^i = \begin{cases} \rho_{Aut}^i + \Delta\rho^i, & \text{positioning} \\ \rho_{Aut}^i + \Delta t \cdot c, & \text{timing} \end{cases} \quad (1)$$

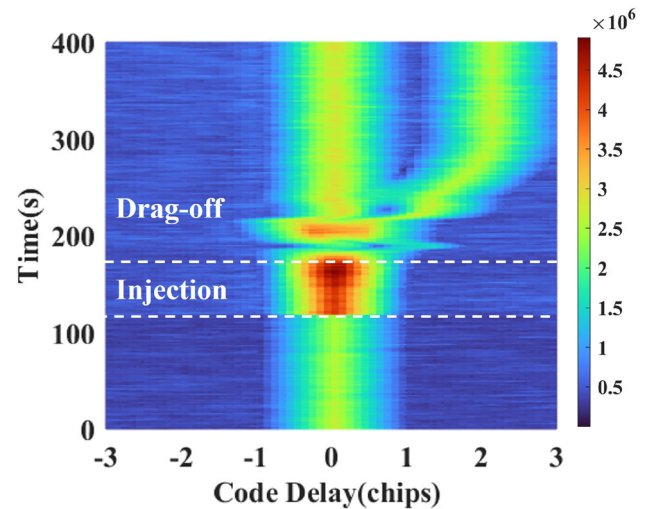


Fig. 1 Time history of receiver correlation results in the code-delay domain under a spoofing attack [21]

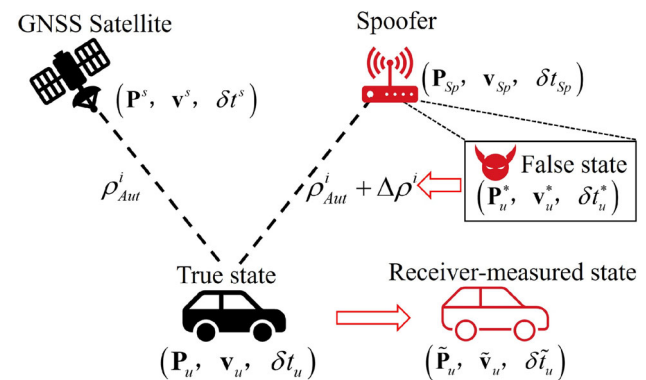


Fig. 2 An example of the spoofing attack

where ρ_0^i is the measured pseudorange from satellite i under attack, ρ_{Aut}^i is the authentic LOS pseudorange, $\Delta\rho^i$ is the spoofed per-channel pseudorange error, Δt is the spoofed receiver-clock bias, and c is the speed of light. For a traditional blind spoofer lacking knowledge of the victim receiver's state, pseudorange biases arise from the baseline between the spoofer and the receiver. These biases can induce geometry-dependent timing or position errors. Moreover, blind spoofers typically require substantially higher power to manipulate or break the receiver's tracking loops and to force re-tracking on the spoofed signals alone. As a result, such spoofers are comparatively easy to detect using power-monitoring methods [4, 32]. Fig. 2 illustrates a typical GNSS spoofing attack scenario, showing the following components: the satellite's PVT ($\mathbf{P}^s, \mathbf{v}^s, \delta t^s$), spoofer's antenna PVT ($\mathbf{P}^{sp}, \mathbf{v}^{sp}, \delta t^{sp}$), the spoofer-target false state ($\mathbf{P}_u^*, \mathbf{v}_u^*, \delta t_u^*$), the user receiver true state ($\mathbf{P}_u, \mathbf{v}_u, \delta t_u$) and the receiver-measured state ($\tilde{\mathbf{P}}_u, \tilde{\mathbf{v}}_u, \tilde{\delta t}_u$) calculated from the pseudorange measurement only.

When the spoofer successfully deceives the victim receiver, the final iteration of the receiver-measured state is expected to be the same as the spoofer-target false state.

The spoofed carrier Doppler can be synthesized in either a *frequency-locked mode*, closely matching the authentic LOS Doppler, or a *frequency-unlocked mode*, in which the spoofer follows its own kinematics and clock. The receiver's per-channel Doppler measurement f_0^i under attack can be written as

$$f_0^i = \begin{cases} f_{Aut}^i, & \text{frequency-locked} \\ f_{Sp}^i, & \text{frequency-unlocked} \end{cases} \quad (2)$$

f_{Aut}^i and f_{Sp}^i are the authentic signal and spoofing signal Doppler frequency, respectively. The frequency-locked or the locked-Doppler mode helps avoid telltale frequency tracking errors during the injection and drag-off phases [30, 33]. This lock may be disabled once the spoofed code phase has been shifted sufficiently far from the true code phase [3]. In practice, maintaining frequency lock can be challenging, especially for a dynamic victim receiver, because the spoofer must know (and track) the geometry between its antenna and the victim's antenna [4]. Nonetheless, it is still imperative to develop resilient defense techniques to prepare a system's vulnerabilities against potential sophisticated spoofing attacks [30, 34].

Alternatively, the attack can operate in a frequency-unlocked mode: the spoofing signal Doppler follows the spoofer's own pseudorange rate, producing a persistent Doppler mismatch relative to the authentic signal. This mismatch can cause loss of lock or frequency tracking errors that reveal the attack. For the receiver loop behavior under frequency-unlocked spoofing, [35] reports a distinctive Doppler error arising from correlations between the spoofed and authentic signals; the error exhibits periodic oscillations until the end of the drag-off phase, offering a direct detection feature.

3 Proposed spoofing detection method

For spoofing detection, this paper introduces the predicted Doppler frequency calculated based on the pseudorange-based navigation solution, which is defined as:

$$f_\rho = \frac{1}{\lambda} \left[(\mathbf{v}^s - \tilde{\mathbf{v}}_u) \tilde{\mathbf{I}} + (\delta \dot{t}_u - \delta \dot{t}^s) c \right] + \tilde{\varepsilon} \quad (3)$$

where λ is carrier wavelength. $\tilde{\mathbf{I}} = \frac{\mathbf{p}^s - \tilde{\mathbf{p}}_u}{\|\mathbf{p}^s - \tilde{\mathbf{p}}_u\|}$ is the unit vector of the signal path from the receiver-measured position to the satellite $\tilde{\mathbf{I}}$ can be assumed to remain unchanged over a short time period due to the long distance between the satellite and

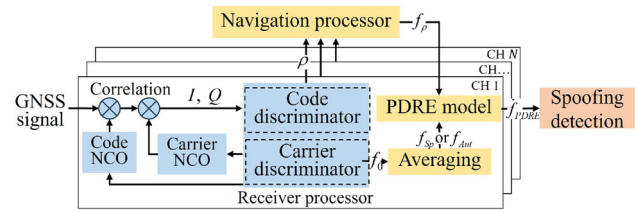


Fig. 3 Architecture of the proposed PDRE algorithm for spoofing detection

receiver. Note that $\tilde{\mathbf{v}}_u$ is computed by taking the derivative of the positioning results in this work. $\delta \dot{t}^s$ and $\delta \dot{t}_u$ are the satellite clock drift and receiver-measured clock drift. Assuming that the satellite clock is stable and the satellite clock drift will not obviously change during a short period. $\tilde{\varepsilon}$ is the noise in the receiver navigation solution, which correlates with signal quality and receiver tracking performance. A detailed discussion on noise will be provided in Section 4. Consequently, the predicted Doppler rate can be written as:

$$\dot{f}_\rho \approx \frac{1}{\lambda} \left[(\mathbf{a}^s - \tilde{\mathbf{a}}_u) \tilde{\mathbf{I}} + \delta \ddot{t}_u c \right] + \dot{\tilde{\varepsilon}} \quad (4)$$

where $\mathbf{a}^s$ and $\tilde{\mathbf{a}}_u$ are the acceleration of the satellite and receiver-measured user receiver, respectively. $\ddot{t}_u$ is the change rate of receiver-measured clock drift, which is non-zero under the timing-based spoofing attacks. The symbol $(\ddot{\cdot})$ denotes the second derivative of the variable. $\dot{\tilde{\varepsilon}}$ is the change rate of the receiver measurement noise. The predicted Doppler rate error (PDRE) is proposed to monitor the inconsistency between the receiver-measured Doppler rate and the predicted Doppler rate:

$$\dot{f}_{PDRE} = \dot{f}_0 - \dot{f}_\rho \quad (5)$$

where f_0 is the receiver Doppler measurement and $\dot{f}_0$ is the Doppler rate derived from f_0 . Spoofing detection is conducted by detecting an abnormality in the PDRE value. Fig. 3 illustrates the architecture of the proposed PDRE model and detection algorithms in the GNSS receiver tracking loop.

Accordingly, the proposed PDRE metric relies solely on pseudorange and Doppler measurements, which are the standard outputs of GNSS receivers, and requires no modifications to the receiver. In the implementation, the predicted Doppler rate is computed by finite differencing the predicted Doppler frequency over consecutive detection windows. The satellite position is first calculated from the decoded broadcast ephemeris, and the satellite velocity is approximated by finite differencing the satellite position over the receiver navigation interval. Similarly, the receiver velocity is obtained by finite differencing consecutive receiver position solutions. In Section 5.2, the receiver navigation interval is set to 1 millisecond (ms), the coherent integration time T is 1 ms, and

the number of samples in each detection window is set to $N = 1000$.

As detailed in Section 2, a spoofer may transmit signals with different Doppler frequency characteristics depending on the selected attack mode, which can affect PDRE performance. This work therefore analyzes the no-spoofing case and four primary spoofing-attack scenarios. The corresponding section references and validation experiments are summarized in Table 1. For clarity, we assume the attack mode remains fixed within each scenario.

3.1 Scenario 1: Spoofing absence

In practical applications, receiver Doppler measurement is influenced by the user receiver clock error, satellite clock error, error due to ionosphere and troposphere delay, and noise. The receiver Doppler measurement in spoofing absence scenarios can be written as:

$$f_0 = f_{Aut} = \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u) \mathbf{I} + (\delta \dot{t}_u - \delta \dot{t}^s) c] - \dot{I}_f + \dot{T}_f + \varepsilon \quad (6)$$

$\mathbf{I} = \frac{\mathbf{P}^s - \mathbf{P}_u}{\|\mathbf{P}^s - \mathbf{P}_u\|}$ is the unit vectors align with the propagation path from the satellite to the receiver's true position. $\delta \dot{t}_u$ is the receiver clock drifts. $\dot{I}_f$ and $\dot{T}_f$ are the time derivatives (rates of change) of the ionospheric and tropospheric delays, respectively. ε denotes the Doppler measurement noise, which is dominated by tracking-loop errors induced by thermal noise (see Section 4). Thus, the time derivative of the Doppler measurement is:

$$\begin{aligned} \dot{f}_0 &= \frac{df_{Aut}}{dt} \\ &= \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I} + (\mathbf{v}^s - \mathbf{v}_u) \dot{\mathbf{I}} + (\delta \ddot{t}_u - \delta \ddot{t}^s) c] + \dot{\varepsilon} \\ &\approx \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I}] + \dot{\varepsilon} \end{aligned} \quad (7)$$

where $\mathbf{a}^s$ and $\mathbf{a}_u$ are the acceleration of the satellite and user receiver, respectively. $\dot{\mathbf{I}}$ is the change rate of the unit vector $\mathbf{I}$, which can be assumed as zero due to the long distance between the satellite and the user receiver. $\delta \ddot{t}_u$ and $\delta \ddot{t}^s$ are the change rates of the user receiver and satellite clock drifts. The satellite and user receiver can be assumed to be driven by the stable clock, and $\delta \ddot{t}_u$ and $\delta \ddot{t}^s$ can be assumed as zero for analytical simplicity. The receiver's short-term oscillator instability may be quantified using Allan deviation or authentic data-based calibration for detection threshold refinement. Second-order temporal variations of the ionosphere and troposphere error are neglected under the assumption that atmospheric delay rates vary slowly within the short detection window considered in this study. In low-earth orbit scenarios

or severe ionospheric disturbance conditions, the second-order atmospheric terms should be explicitly modeled [36].

In the absence of spoofing, multipath reception may still occur. In urban environments, reflected signal components typically exhibit reduced power and high temporal dynamics, which can induce unstable errors in the receiver's PVT solution. Multipath-induced positioning errors are commonly on the order of tens of meters and vary rapidly [37, 38]. Over a relatively long detection window (e.g., 1 second), the resulting position estimates can be treated as the receiver true states relative to spoofing interference. Therefore, in the spoofing-free scenario the receiver's PVT estimate can be modeled as the true state for detection purposes ($\tilde{\mathbf{I}} \approx \mathbf{I}$, $\tilde{\mathbf{v}}_u \approx \mathbf{v}_u$, $\delta \dot{t}_u \approx \delta \dot{t}_u$). The predicted Doppler and Doppler rate in (3) and (4) can be rewritten as:

$$f_\rho = \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u) \mathbf{I} + (\delta \dot{t}_u - \delta \dot{t}^s) c] + \tilde{\varepsilon} \quad (8)$$

$$\dot{f}_\rho \approx \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I}] + \dot{\tilde{\varepsilon}} \quad (9)$$

Substitute (7) and (9) into (5), the PDRE consists of noise only

$$\dot{f}_{PDRE} = \dot{\varepsilon} - \dot{\tilde{\varepsilon}} \quad (10)$$

3.2 Scenarios 2-5: spoofing presence

During the drag-off period of the spoofing attack, the spoofing signal have greater power than the authentic signal to take over the tracking loop. Therefore, the Doppler measurement in the spoofing presence case can be written as the spoofing signal Doppler frequency and generally written as:

$$\begin{aligned} f_0 &= f_{Sp} \\ &= \underbrace{\frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u^*) \mathbf{I}^* + (\delta \dot{t}_u^* - \delta \dot{t}^s) c]}_{f_{Aut}^*} \\ &\quad + \underbrace{\frac{1}{\lambda} [(\mathbf{v}_{Sp} - \mathbf{v}_u) \mathbf{I}_{Sp} + (\delta \dot{t}_u - \delta \dot{t}_{Sp}) c]}_{f_{dyn}} + \varepsilon \end{aligned} \quad (11)$$

where f_{Aut}^* indicates the spoofer-target authentic signal Doppler frequency generated based on the spoofer-target false state. Regarding the difficulty of precisely predicting the victim receiver's velocity in real-world spoofing attacks, the Doppler bias f_{dyn} due to the relative motion between the spoofer and the victim receiver is considered in the receiver-dynamic case. $\mathbf{I}^* = \frac{\mathbf{P}^s - \mathbf{P}_u^*}{\|\mathbf{P}^s - \mathbf{P}_u^*\|}$ is the unit vectors aligned with the propagation paths from the satellite to the from the target false position to the satellite. $\delta \dot{t}_u^*$ is spoofed user clock drift.

Table 1 Spoofing attack scenarios

Spoofing Attack Scenarios	Theoretical Analysis	Experimental Validation
Scenario 1: spoofing absence	Section 3.1	TEXBAT clean-static; TEXBAT clean-dynamic
Scenario 2: spoof time in frequency-locked mode	Section 3.2.1	TEXBAT ds3
Scenario 3: spoof time in frequency-unlocked mode	Section 3.2.2	TEXBAT ds2; TEXBAT ds5
Scenario 4: spoof position in frequency-locked mode	Section 3.2.3	TEXBAT ds4; TEXBAT ds6
Scenario 5: spoof position in frequency-unlocked mode	Section 3.2.4	Simulation test

$\mathbf{I}_{Sp} = \frac{\mathbf{P}_{Sp} - \mathbf{P}_u}{\|\mathbf{P}_{Sp} - \mathbf{P}_u\|}$ is the unit vector aligned with the propagation paths from the victim receiver's true position to the spoofer. δi_{Sp} is the spoofer clock drift.

This paper discusses spoofing attack modes of two groups: frequency-locked mode and frequency-unlocked mode. The first type of spoofers operate in controlled laboratory conditions, with accurate information about the authentic signal, and they are able to predict and mimic the authentic signal Doppler frequency ($f_{Aut}^* = f_{Aut}$, $f_{dyn} = 0$). The second type spoofers, with limited victim receiver velocity information, generate carrier frequency based on the spoofer-target false state.

3.2.1 Scenario 2: spoof time in frequency-locked mode

The receiver timing is spoofed in this scenario. The pseudoranges of spoofing signals are designed to shift based on the desired receiver clock error. Therefore, the pseudorange differences between the spoofing and authentic signals are the same for all satellites. At the same time, with the knowledge of the precise victim receiver PVT information, the spoofer-victim relative geometry is considered. Hence, the spoofer is able to maintain approximately the same Doppler frequency with the authentic signal, written as ($\mathbf{I}^* \approx \mathbf{I}$, $\mathbf{v}_u^* \approx \mathbf{v}_u$, $\delta i_u^* \approx \delta i_u$, $f_{dyn} \approx 0$). Consequently, (11) and its rate of change can be written as:

$$f_0 = f_{Sp} = \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u) \mathbf{I} + (\delta i_u - \delta i^s) c] + \varepsilon \quad (12)$$

$$\begin{aligned} \dot{f}_0 &= \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I} + (\mathbf{v}^s - \mathbf{v}_u) \dot{\mathbf{I}} + (\delta \ddot{i}_u - \delta \ddot{i}^s) c] + \dot{\varepsilon} \\ &\approx \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I}] + \dot{\varepsilon} \end{aligned} \quad (13)$$

Therefore, in this scenario, f_{Sp} is the same with f_{Aut} due to the frequency-locked mode.

Assume that the spoofer successfully drifts the victim receiver timing solution while the velocity and position solutions remain unbiased. The receiver-measured state can be written as: ($\tilde{\mathbf{I}} \approx \mathbf{I}$, $\tilde{\mathbf{v}}_u \approx \mathbf{v}_u$, $\delta \tilde{i}_u \approx \delta i_u^*$). The predicted Doppler and Doppler rate in (3) and (4) can be rewritten

as:

$$f_\rho = \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u) \mathbf{I} + (\delta i_u^* - \delta i^s) c] + \tilde{\varepsilon} \quad (14)$$

$$\dot{f}_\rho \approx \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I} + \delta \ddot{i}_u^* c] + \dot{\tilde{\varepsilon}} \quad (15)$$

Substitute (13) and (15) into (5), the PDRE models for receiver-static and receiver-dynamic cases are the same:

$$\dot{f}_{PDRE} = -\frac{\delta \ddot{i}_u^* c}{\lambda} + \dot{\varepsilon} - \dot{\tilde{\varepsilon}} \quad (16)$$

The discrepancy between the predicted and receiver-measured Doppler rate primarily originates from spoofer-induced timing errors. In practical attacks that are not perfectly frequency-locked, the dynamic-frequency component f_{dyn} introduces a systematic (nonzero-mean) bias in the PDRE, providing stronger evidence for the presence of spoofing signals.

3.2.2 Scenario 3: spoof time in frequency-unlocked mode

In this scenario, the spoofer typically generates the carrier wave based on the spoofer-target false state. Therefore, the Doppler frequencies of spoofing signals shift equally for different satellites based on the spoofer-target receiver clock error ($\mathbf{v}_u^* \approx \mathbf{v}_u$, $\mathbf{I}^* \approx \mathbf{I}$, $\delta i_u^* \approx \delta i_u^*$). (11) and its rate of change for the victim receiver-static (Static) and receiver-dynamic (Dynamic) cases can be written as:

$$f_0 = f_{Sp} = \begin{cases} \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u) \mathbf{I} + (\delta i_u^* - \delta i^s) c] + \varepsilon & , \text{Static} \\ \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u) \mathbf{I} + (\delta i_u^* - \delta i^s) c] + \frac{1}{\lambda} [(-\mathbf{v}_u) \mathbf{I}_{Sp} + (\delta i_u - \delta i_{Sp}) c] + \varepsilon & , \text{Dynamic} \end{cases} \quad (17)$$

$$\dot{f}_0 \approx \begin{cases} \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I} + \delta \ddot{i}_u^* c] + \dot{\varepsilon} & , \text{Static} \\ \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I} + \delta \ddot{i}_u^* c] + \frac{1}{\lambda} [-\mathbf{a}_u \mathbf{I}_{Sp} - \mathbf{v}_u \dot{\mathbf{I}}_{Sp}] + \dot{\varepsilon} & , \text{Dynamic} \end{cases} \quad (18)$$

In this paper, it is assumed that the spoofer always remains stationary ($\mathbf{v}_{Sp} = 0$). $\dot{\mathbf{I}}_{Sp}$ is the change rate of the unit vector $\mathbf{I}_{Sp}$. The receiver-measured state is the same as the spoofer-target state, which is the same in scenario 2. Therefore, the

predicted Doppler frequency and Doppler rate are the same as (14) and (15). Consequently, the PDRE for receiver-static and receiver-dynamic cases are:

$$\dot{f}_{PDRE} = \begin{cases} \dot{\varepsilon} - \ddot{\varepsilon} \\ \frac{1}{\lambda} (-\mathbf{a}_u \mathbf{I}_{Sp} - \mathbf{v}_u \dot{\mathbf{I}}_{Sp}) + \dot{\varepsilon} - \ddot{\varepsilon} \end{cases}, \text{Static/Dynamic} \quad (19)$$

Abnormal PDRE bias arises from the relative motion between the spoofer and the victim receiver in dynamic scenarios. For a single-antenna spoofer, the PDRE bias tends to be consistent across channels, whereas for a multi-antenna spoofer it can vary from channel to channel. In the receiver-static case, PDRE may be less discriminative and can fail to indicate an ongoing spoofing attack. However, during the spoofing-signal injection, a Doppler mismatch often produces noticeable fluctuations in the C/N_0 and in the carrier-frequency estimate $\tilde{f}_0$ [3]. Additionally, for attacks operating in the frequency-unlocked mode, the Doppler difference between the spoofed and authentic signals induces a characteristic oscillation that can be readily detected by prior work leveraging Doppler-oscillation features [35].

3.2.3 Scenario 4: spoof position in frequency-locked mode

In this scenario, the spoofer generates signals with a drifted pseudorange based on the false position. Under frequency-locked mode, the measured Doppler frequency and Doppler rate are the same as (12) and (13). Assume that the spoofer has successfully spoofed the user's position and maintained an unbiased clock drift. The receiver-measured state can be written as: $(\tilde{\mathbf{I}} \approx \mathbf{I}^*, \tilde{\mathbf{v}}_u \approx \mathbf{v}_u^*, \delta \tilde{t}_u \approx \delta t_u)$. Predicted Doppler frequency and Doppler rate are:

$$f_\rho = \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u^*) \mathbf{I}^* + (\delta \dot{t}_u - \delta \dot{t}^s) c] + \tilde{\varepsilon} \quad (20)$$

$$\dot{f}_\rho \approx \frac{1}{\lambda} (\mathbf{a}^s - \mathbf{a}_u^*) \mathbf{I}^* + \dot{\tilde{\varepsilon}} \quad (21)$$

The PDRE models for receiver-static and receiver-dynamic cases are the same:

$$\dot{f}_{PDRE} = \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u) \mathbf{I} - (\mathbf{a}^s - \mathbf{a}_u^*) \mathbf{I}^*] + \dot{\varepsilon} - \ddot{\varepsilon}$$

The PDRE bias is driven by the mismatch between the receiver's true state and the spoofer-target (false) state. As this mismatch increases, the PDRE bias grows, thereby enhancing spoofing detectability.

3.2.4 Scenario 5: spoof position in frequency-unlocked mode

The spoofer generates the carrier according to the false position, the shift in Doppler frequency varying for different satellites $(\mathbf{I}^* \approx \mathbf{I}^*, \mathbf{v}_u^* \approx \mathbf{v}_u^*, \delta \dot{t}_u^* \approx \delta \dot{t}_u)$. Therefore, the

receiver-measured Doppler and Doppler rate of spoofing signals can be modeled as:

$$f_0 = f_{Sp} = \begin{cases} \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u^*) \mathbf{I}^* + (\delta \dot{t}_u - \delta \dot{t}^s) c] + \varepsilon, \text{Static} \\ \frac{1}{\lambda} [(\mathbf{v}^s - \mathbf{v}_u^*) \mathbf{I}^* + (\delta \dot{t}_u - \delta \dot{t}^s) c] + \varepsilon, \text{Dynamic} \end{cases} \quad (22)$$

$$\dot{f}_0 \approx \begin{cases} \frac{1}{\lambda} (\mathbf{a}^s - \mathbf{a}_u^*) \mathbf{I}^* + \dot{\varepsilon}, \text{Static} \\ \frac{1}{\lambda} [(\mathbf{a}^s - \mathbf{a}_u^*) \mathbf{I}^* - \mathbf{a}_u \mathbf{I}_{Sp} - \mathbf{v}_u \dot{\mathbf{I}}_{Sp}] + \dot{\varepsilon}, \text{Dynamic} \end{cases} \quad (23)$$

Similar to scenario 4, the spoofer successfully spoofed the user position $(\tilde{\mathbf{I}} \approx \mathbf{I}^*, \tilde{\mathbf{v}}_u \approx \mathbf{v}_u^*, \delta \tilde{t}_u \approx \delta t_u)$. The predicted Doppler frequency and Doppler rate are the same as (20) and (21). Hence, the PDRE is:

$$\dot{f}_{PDRE} = \begin{cases} \dot{\varepsilon} - \ddot{\varepsilon} \\ \frac{1}{\lambda} (-\mathbf{a}_u \mathbf{I}_{Sp} - \mathbf{v}_u \dot{\mathbf{I}}_{Sp}) + \dot{\varepsilon} - \ddot{\varepsilon} \end{cases}, \text{Static/Dynamic} \quad (24)$$

In conclusion, the proposed PDRE method demonstrates theoretical effectiveness across most spoofing attack scenarios, as outlined in Table 2. The theoretical analysis also applies to the traditional overwhelming spoofer with the same attack mode. Spoofing attack scenarios 2 to 5 correspond to the spoofing attack modes detailed in Table 1. Table 2 further categorizes these scenarios into a total of 16 cases based on user dynamics and the number of spoofing antennas. 'Yes' indicates the PDRE is effective in detecting a spoofing attack, while 'No' denotes theoretical ineffectiveness.

The proposed metric demonstrates the capability to continuous spoofing detection, even in cases where spoofers employ switching attack modes, such as transitioning between drift positioning and timing, or from frequency-locked to frequency-unlocked modes. This feature is especially advantageous in receiver-dynamic applications.

4 Detection algorithm

The performance of the PDRE in the absence and presence of the spoofing attack can be described as a binary hypothesis test. The likelihood ratio test (LRT) is used for spoofing detection [39]. The null hypothesis H_0 indicates the absence of a spoofing signal, while the alternative hypothesis H_1 indicates the presence of the spoofing attack:

$$\begin{aligned} H_0 : y &= n \sim \mathcal{N}(0, \sigma^2) \\ H_1 : y &= b + n \sim \mathcal{N}(b, \sigma^2) \end{aligned} \quad (25)$$

where y indicates the PDRE value. n is noise, which consists of noise from the receiver Doppler measurement and the predicted Doppler frequency. n can be modeled by the additive complex Gaussian distribution with zero mean and a variance of σ^2 . b is the PDRE bias due to a spoofing attack. Assuming

Table 2 Theoretical detection effectiveness of the proposed PDRE method under different spoofing scenarios

Victim receiver	Spoofing configuration	Spoofing scenario			
		Scenario 2	Scenario 3	Scenario 4	Scenario 5
User-static	Single-antenna spoofer	Yes	No	Yes	No
	Multi-antenna spoofer	No	No	Yes	No
User-dynamic	Single-antenna spoofer	Yes	Yes	Yes	Yes
	Multi-antenna spoofer	Yes	Yes	Yes	Yes

that the dominant sources of receiver Doppler measurement noise are the GNSS receiver frequency-locked loop (FLL) tracking error, deriving from thermal noise and dynamic stress error. This paper does not consider dynamic stress error due to variations in modeling across different receiver loops and the assumption of utilizing a higher-order FLL. Therefore, the receiver Doppler measurement noise can be modeled by the additive complex Gaussian distribution with zero mean and a variance of σ_{iFLL}^2 . σ_{iFLL} can be estimated based on the tracking loop parameters and signal power, as illustrated in Appendix.A. The variance of the averaged Doppler measurement during a detection window with a size of N samples is as follows:

$$\text{var}(\varepsilon) = \frac{\sigma_{iFLL}^2}{N} \quad (26)$$

The Doppler rate is estimated from the temporal variation of the average Doppler measurements. For two consecutive non-overlapping detection windows, each with a duration of NT , the Doppler-rate error at epoch k can be approximated by the first-order finite difference

$$\dot{\varepsilon}_k \approx \frac{\varepsilon_{k+1} - \varepsilon_k}{NT} \quad (27)$$

where T denotes the time interval between two consecutive Doppler measurements. Accordingly, the variance of the Doppler rate is

$$\text{var}(\dot{\varepsilon}_k) = \text{var}\left(\frac{\varepsilon_{k+1} - \varepsilon_k}{NT}\right) = \frac{1}{N^2 T^2} \text{var}(\varepsilon_{k+1} - \varepsilon_k) \quad (28)$$

According to (26),

$$\text{var}(\varepsilon_k) = \text{var}(\varepsilon_{k+1}) = \frac{\sigma_{iFLL}^2}{N} \quad (29)$$

For two non-overlapping windows, the averaged Doppler errors ε_k and ε_{k+1} can be assumed independent. Therefore,

$$\text{var}(\varepsilon_{k+1} - \varepsilon_k) = \text{var}(\varepsilon_{k+1}) + \text{var}(\varepsilon_k) \quad (30)$$

Hence,

$$\text{var}(\dot{\varepsilon}_k) = \frac{2\sigma_{iFLL}^2}{N^3 T^2} \quad (31)$$

Predicted Doppler noise mainly results from the noise in the receiver positioning solution. The predicted Doppler frequency noise is related to the pseudorange-rate $\dot{\rho}$ noise by

$$f_\rho = \frac{1}{\lambda} \dot{\rho} \quad (32)$$

The variance of the pseudorange-rate error $\dot{\varepsilon}_\rho$ is given by

$$\text{var}(\dot{\varepsilon}_{\rho,k}) = \text{var}\left(\frac{\varepsilon_{\rho,k+1} - \varepsilon_{\rho,k}}{T}\right) \quad (33)$$

ε_ρ is the pseudorange error due to thermal noise, which can be estimated by the delay-locked loop (DLL) tracking jitter model illustrated in Appendix.A. Similar to (30) and according to (A.3), the variance of the predicted Doppler frequency noise $\tilde{\varepsilon}$ can be written as

$$\text{var}(\tilde{\varepsilon}) = \frac{1}{\lambda^2} \text{var}(\dot{\varepsilon}_{\rho,k}) = \frac{2\sigma_{iDLL}^2 \lambda_c^2}{\lambda^2 T^2} = \sigma_{PD}^2 \quad (34)$$

Similar to the derivations from (27) to (31), the variance of the predicted Doppler frequency rate after averaging is

$$\text{var}\left(\dot{\tilde{\varepsilon}}\right) = \frac{2\sigma_{PD}^2}{N^3 T^2} \quad (35)$$

The receiver-measured Doppler rate and predicted Doppler frequency rate are independent, and the variance of PDRE is:

$$\text{var}(\dot{f}_{PDRE}) = \text{var}(\dot{\varepsilon}) + \text{var}\left(\dot{\tilde{\varepsilon}}\right) = \frac{2(\sigma_{iFLL}^2 + \sigma_{PD}^2)}{N^3 T^2} = \sigma^2 \quad (36)$$

Ultimately, the probability density functions (PDFs) of the PDRE values are

$$\begin{aligned} p(y|H_0) &= \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left[-\frac{1}{2\sigma^2} y^2\right] \\ p(y|H_1) &= \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left[-\frac{1}{2\sigma^2} (y-b)^2\right] \end{aligned} \quad (37)$$

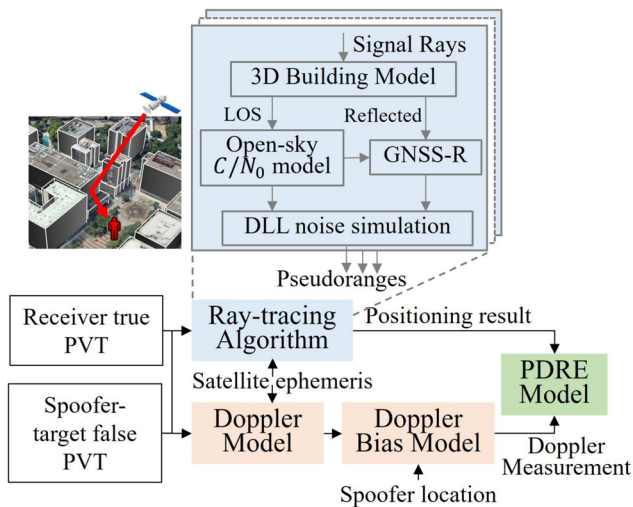


Fig. 4 Architecture of the proposed spoofing attack simulator

The Neyman-Pearson (NP) approach decides H_1 if the likelihood ratio is larger than a threshold.

$$L(y) = \frac{p(y|H_1)}{p(y|H_0)} > \gamma \quad (38)$$

The test statistic can be written as $T(y) = y$. Consequently, the detection probability P_D and false alarm probability P_{FA} are

$$\begin{aligned} P_D &= \Pr\{y > \gamma'; H_1\} \\ P_{FA} &= \Pr\{y > \gamma'; H_0\} \end{aligned} \quad (39)$$

The threshold γ' can be adjusted to control P_{FA} .

5 Experiment

5.1 Simulation tests

A spoofing-attack simulator is developed to reproduce realistic receiver positioning, Doppler-measurement behavior, and PDRE detection performance. Assisted by ray-tracing, the simulator accounts for the victim receiver's surrounding environment and multipath interference. The overall architecture is shown in Fig. 4. The algorithm comprises three modules: Doppler measurements, navigation solution, and PDRE detection.

First, the receiver's true state and the spoofer's target (false) state are specified, and satellite ephemerides are used to predict potential signal paths. In urban areas, signal propagation is classified as LOS, multipath, or non-line-of-sight (NLOS). Satellite visibility is determined from the receiver's true state using a three-dimensional (3D) building model. The spoofing signal Doppler is then generated according

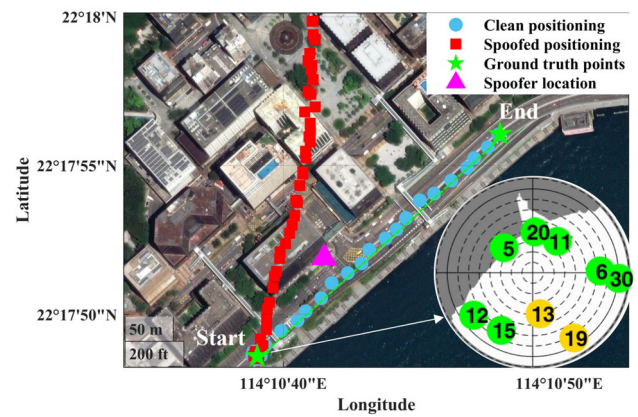


Fig. 5 Spoofing attack scenario, simulated receiver positioning results and the skyplot with visible satellite. Green PRN for LOS type, yellow for multipath type. GPST from 538201s

to the satellite to false state geometry, while the Doppler bias arising from the spoofer-victim relative motion is computed from the predefined static spoofer location and the receiver's true state. Based on (11), combining these yields the noise-free Doppler measurement of the receiver under spoofing. Second, the receiver navigation solution is estimated considering multipath interference on the pseudorange measurement. Using ray-tracing, the dominant reflected path and its code delay are obtained. The C/N_0 of the authentic LOS signal is simulated via an elevation- C/N_0 regression model, and the reflected signal C/N_0 is computed assuming specular reflection with attenuation given by reflection coefficients from a GNSS reflectometry (GNSS-R) model. Details of the ray-tracing procedure follow [40, 41]. Assuming an early-minus-late power discriminator, multipath error on the pseudorange is generated from the standard multipath error envelope [1], and pseudorange noise is added according to code-tracking jitter due to thermal noise. This yields realistic urban positioning errors. Finally, the proposed PDRE model is applied for spoofing detection. The simulator also enables evaluation of PDRE false alarms under multipath and noisy conditions.

The simulation experiments are carried out to assess the detection performance of PDRE on a common spoofing attack scenario aimed at manipulating a vehicle's positioning in frequency-unlocked mode, which corresponds to scenario 5 in Section 3.2.4. Fig. 5 illustrates the simulation environment, sky-plot, victim receiver true location, spoofer location, simulated receiver positioning result in spoofing absence (clean) case, receiver positioning results under spoofing presence (spoofed) on July 15, 2023, and Global Positioning System time (GPST) from 538201. According to ray-tracing classification results of the signal rays, PRNs 13 and 19 are under multipath interference.

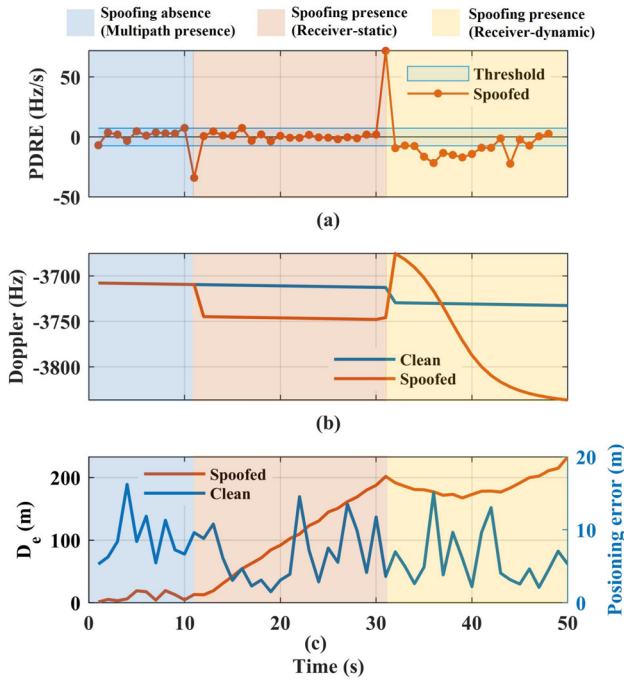


Fig. 6 Time history of (a) PDRE value, (b) Doppler frequency of PRN 19, and (c) D_e under spoofing presence and positioning error under spoofing absence cases, with $P_{FA} = 1\%$

In this test, the DLL noise bandwidth is set to 0.2 Hz and the early-to-late correlator spacing is one chip. The victim receiver is set to remain static at the start point for 30 seconds and move to the endpoint at a constant speed of around 18 meters per second (m/s). The spoofing attack begins at 11 seconds and results in a graduated deviation in receiver positioning. Therefore, from 0 to 10 seconds corresponds to the spoofing absence case, from 11 to 30 seconds simulates the spoofing attack scenario 5 with static victim receiver, and from 30 to 50 seconds simulates the dynamic victim receiver under spoofing attack scenario 5. Assume that the final iteration of the receiver-measured state is the same as the spoofer-target false state. Define $D_e = P_u - P_u^*$ as the vector of the position error between the false and true states, to indicate the spoofing attack process. Positioning error represents the difference between the simulated receiver positioning and the receiver ground true location in spoofing absence case, reflecting the potential multipath or noise interference. Fig. 6 is the time history of PDRE value with detection threshold, Doppler frequency of PRN 19 under clean and spoofed cases, D_e under spoofed case, and positioning error under clean case.

The result shows that the spoofing attack leads to a constant Doppler frequency error from 11 seconds. PDRE is bound by the threshold until a notable deviation occurs at 31 seconds, when the victim receiver starts to move. The PDRE behavior is consistent with the theoretical analysis, which

successfully detects spoofing in the receiver-dynamic case of spoofing attack scenario 5. However, the PDRE detection sensitivity is degraded in the last 10 seconds, probably due to the growing distance between the spoofer and the victim receiver. Note that the positioning error due to multipath or noise interference is around 10 meters and does not lead to a false alarm in the first 10 seconds.

This section then conducts simulation experiments to analyze the factors that influence PDRE detection performance, including detection window length, measurement noise level, the spoofer's target false state, and the victim receiver's dynamics. The simulation environment and spoofer locations are identical to those shown in Fig. 5. The victim receiver is assumed to move at constant velocity (zero acceleration) throughout the simulation. For clarity, the C/N_0 is set to the same value for all satellites. The DLL and FLL noise bandwidths are 1.5 Hz and 10 Hz, respectively. In the first simulation scenario, the spoofer operates in a frequency-locked mode, so the PDRE follows the theoretical model introduced in spoofing attack scenario 4 in Section 3.2.3. Fig. 7(a) reports detection performance versus detection window length and noise level, while Fig. 7(b) examines the impact of the spoofer-target false state. P_D and P_{FA} are the probability of detection and probability of false alarm, respectively. Define the acceleration-state error as $a_e = a_u - a_u^*$ where a_u is the receiver's true acceleration and a_u^* is the spoofer-target acceleration.

The results indicate that the proposed PDRE achieves better detection performance with a larger window length N and lower measurement noise. Under highly dynamic motion, however, N must balance accuracy against sensitivity: a larger N suppresses noise more effectively but can attenuate short-lived biases. We further observe that the acceleration error a_e strongly influences the PDRE, whereas the position error D_e has little effect. This is because of the long distance between satellites and the false state, resulting in minimal alteration in the signal vector during a spoofing attack. It is noteworthy that the PDRE value depends on the projection of a_e onto the signal path. Therefore, the PDRE detection performance correlates with the direction of spoofer-target false states. Generally, under a fixed signal path, a greater acceleration error between false and true states corresponds to a higher probability of spoofing detection. In practice, it is possible for the receiver-measured positioning and acceleration errors to reach significant magnitudes due to tracking errors during spoofing signal injection. In such cases, the receiver-measured state differs from the spoofer target false state. Nonetheless, these spoofer-unexpected positioning and acceleration errors can enhance the effectiveness of PDRE for detecting spoofing.

Fig. 8 depicts the detection performance of PDRE in a scenario where the spoofer is in frequency-unlocked mode, aligning with the theoretical analysis outlined in Section

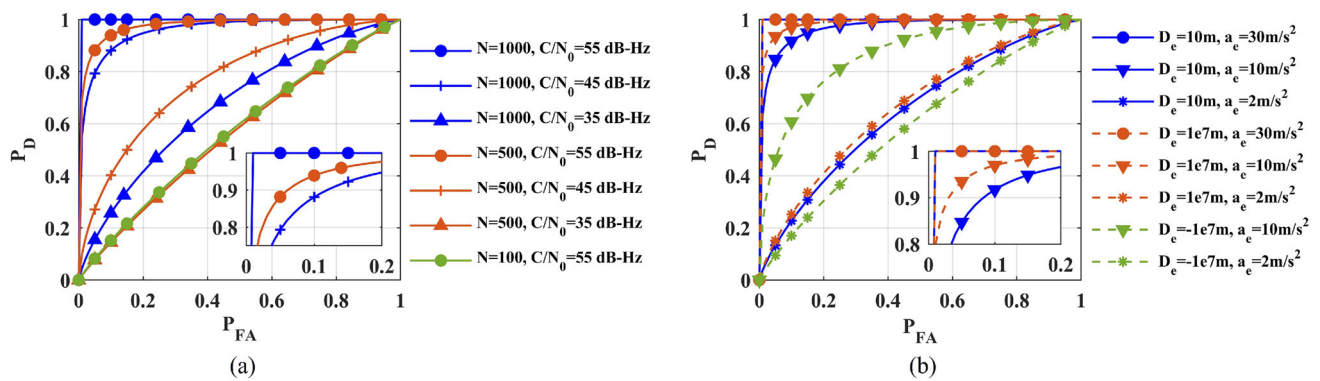


Fig. 7 (a) Effect of N and C/N_0 value with $D_e = 100\text{m}$ and $a_e = 30\text{m/s}^2$, and (b) effect of D_e and a_e value with $N = 1000$ and $C/N_0 = 55\text{dB-Hz}$ for spoofing attack mode in positioning-based and frequency-locked mode

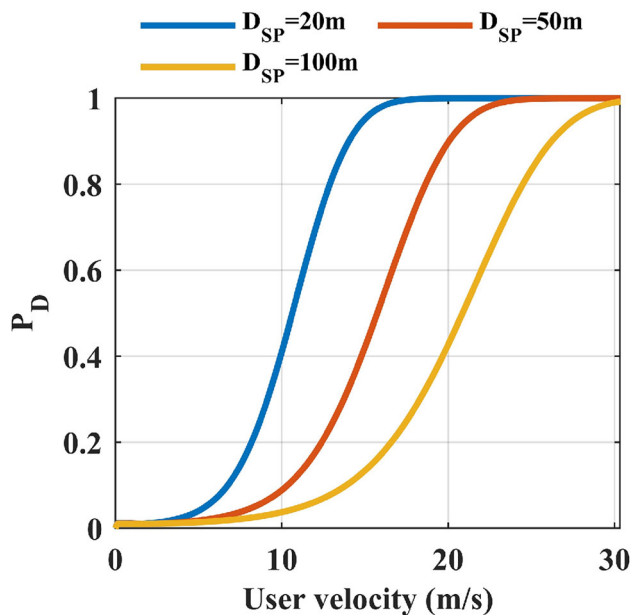


Fig. 8 Detection performance with different user velocity and spoofer location for the spoofing attack in positioning-based and frequency-unlocked mode with $N = 1000$ and $C/N_0 = 55\text{dB-Hz}$, with $P_{FA} = 1\%$

3.2.4. $D_{SP} = \mathbf{P}_u - \mathbf{P}_{Sp}$ is defined as the vector of the distance between the victim receiver and spoofer antennas. The PDRE value primarily derives from the component $\mathbf{v}_u \cdot \hat{\mathbf{I}}_{SP}$, given the victim receiver maintains a constant velocity throughout this simulation. Note that the detection performance varies based on the projection of the user velocity onto the vector $\hat{\mathbf{I}}_{SP}$. Holding other impact factors constant, a shorter distance D_{SP} corresponds to a larger $\hat{\mathbf{I}}_{SP}$, consequently leading to a higher probability of spoofing detection. Therefore, the proposed PDRE method offers significant advantages for vehicular applications.

5.2 TEXBAT dataset

This section evaluates the spoofing detection performance using the widely used and publicly available Texas Spoofing Test Battery (TEXBAT) dataset, and detailed descriptions of the attack modes can be found in [30]. The TEXBAT dataset can be used for evaluating detection performance in advanced synchronous spoofing attacks, where both authentic and spoofing signals can be received by receivers during the beginning of the attack. There are five spoofing attack scenarios in the TEXBAT dataset corresponding to ds2, ds3, ds4, ds5 and ds6. In these scenarios, the frequency-locked mode is achieved with precise knowledge of the victim receiver's PVT information, while the frequency-unlocked mode is achieved by shifting the carrier phase proportional to the code phase. Hence, the spoofing attack mode in ds2 aligns with the receiver-static case in scenario 3, as defined in Section 3.2. Similarly, ds3 corresponds to the receiver-static case in scenario 2, ds4 corresponds to the receiver-static case in scenario 4, and ds6 corresponds to the receiver-dynamic case in scenario 4. However, ds5 of the TEXBAT dataset operates in a receiver-dynamic and frequency-unlocked mode, where the relative motion between the spoofer and victim receiver may not be considered, as the spoofing signal is predicted and directly combined with the antenna output of the victim receiver. Consequently, the proposed PDRE method is theoretically ineffective in ds5.

The clean datasets of TEXBAT serve to indicate the true states of the victim receiver and receiver measurements in the absence of the spoofing signal. Therefore, the time history of D_e , a_e , and the receiver timing error $\delta t_e = \delta t_u - \delta t_e^*$ can be used to indicate spoofing attack procedures. Fig. 9(a) presents the PDRE values and receiver Doppler measurements of PRN 23 in both clean static data and TEXBAT ds2, as well as the magnitudes of D_e , a_e and δt_e of TEXBAT ds2. The results indicate that the spoofed PDRE closely resembles the clean PDRE, suggesting that the spoofed PDRE comprises primar-

ily noise. The receiver Doppler measurement and clock error reveal that the spoofer begins to manipulate the receiver clock measurement around 150s. Furthermore, the spoofing attack induces approximately a 10m positioning error starting from around 100s. In conclusion, the proposed PDRE is unable to detect spoofing attacks in the static case of spoofing attack scenario 3, which is consistent with the theoretical analysis summarized in Table 2.

Fig.9 (b) and (c) depict the time history of the PDRE, receiver Doppler measurement, D_e , a_e and δt_e for TEXTBAT ds3 and ds4, respectively. The performance of spoofed PDRE is correlated to the rise in positioning and receiver clock error. Compared to the clean PDRE, the spoofed PDRE in TEXTBAT ds2 and ds3 exhibit a distinctive increase in variance from around 150s, indicating the presence of the spoofing attack. For receiver-dynamic cases shown in Fig.10, the spoofed PDRE exhibits a large deviation from the clean PDRE, indicating an effective spoofing detection ability.

Notably, it is observed that a timing-only spoofing attack may also induce receiver positioning inaccuracies, and a similar phenomenon is observed in positioning-only spoofing attacks. This phenomenon may result from the unforeseen correlation between the LOS and spoofing signals within the receiver tracking loop or settings within the navigation processor of the GNSS receiver. Additionally, missing or erroneous predictions of f_{dyn} may exist in the generation of the spoofing signal. Nevertheless, the unexpected receiver-measured state error or receiver Doppler measurement error will introduce bias in the PDRE, further enhancing the PDRE method for spoofing detection. For example, the proposed PDRE method remains effective in indicating the presence of spoofing in TEXTBAT ds5.

To distinguish a spoofing attack from signal degradation caused by multipath interference or natural fading, a moving threshold based on the C/N_0 is employed for detection. The estimation of the threshold following the theoretical deviation in Section 4. The data is post-processed using a MATLAB-based GPS software-defined receiver [42]. The DLL noise bandwidth is set to 1.5 Hz, and the FLL noise bandwidth is 10 Hz. Typically, a lower C/N_0 results in a higher threshold, influencing the sensitivity of the proposed PDRE for spoofing detection. Fig.11 illustrates the time history of spoofed PDRE value alongside the corresponding threshold values in TEXTBAT ds2, ds4, and ds6.

In TEXTBAT ds2, the spoofed PDRE is buried within the noise floor and is submerged beneath the threshold. Therefore, no spoofing attack is detected. In TEXTBAT ds4, the PDRE successfully signals the spoofing attack from around 110s, corresponding to the spoofing injection stage, and exhibits more frequent and consistent alerts from approximately 180 seconds onwards, indicating the spoofer's drag-off stage. A single false alarm occurred during the first epoch of the test, likely caused by the receiver's unstable tracking

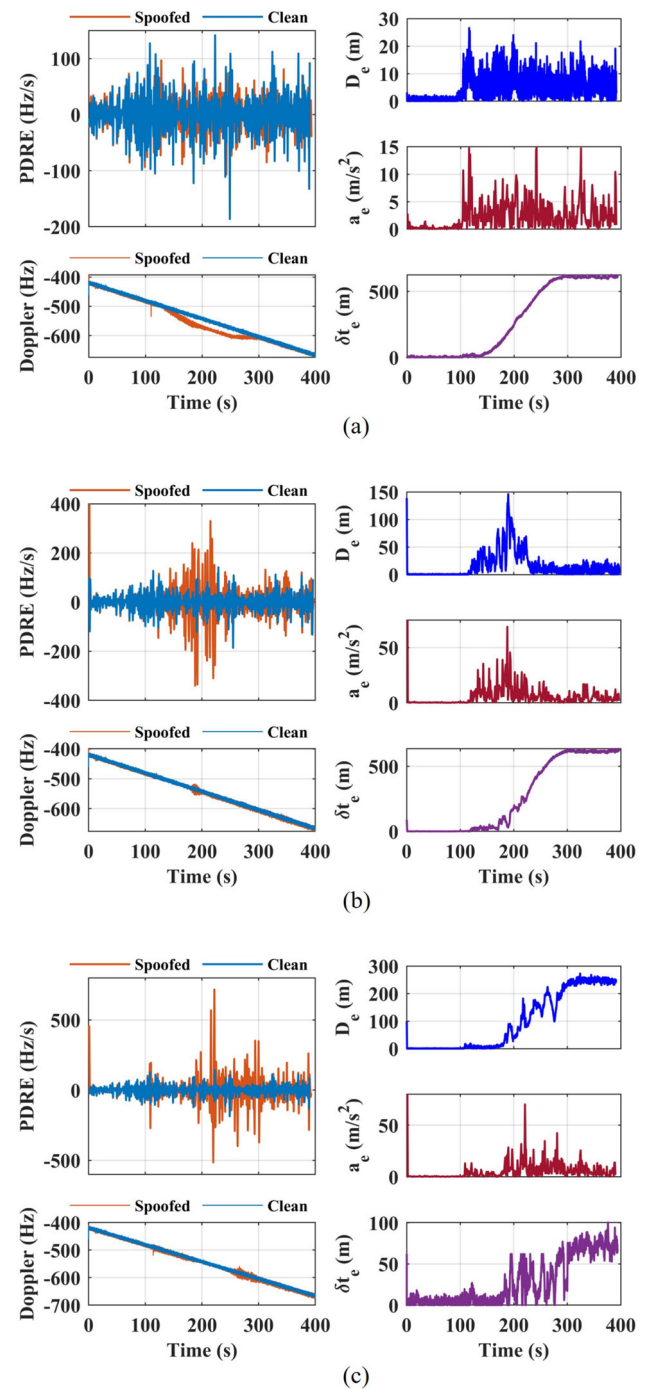


Fig. 9 PDRE and receiver-measured Doppler frequency obtained in the spoofing absence (clean) data and spoofing presence (spoofed) data, the magnitude of positioning error, acceleration error, and receiver clock error due to spoofing attack in PRN 23 of (a) TEXTBAT ds2, (b) TEXTBAT ds3 and (c) TEXTBAT ds4

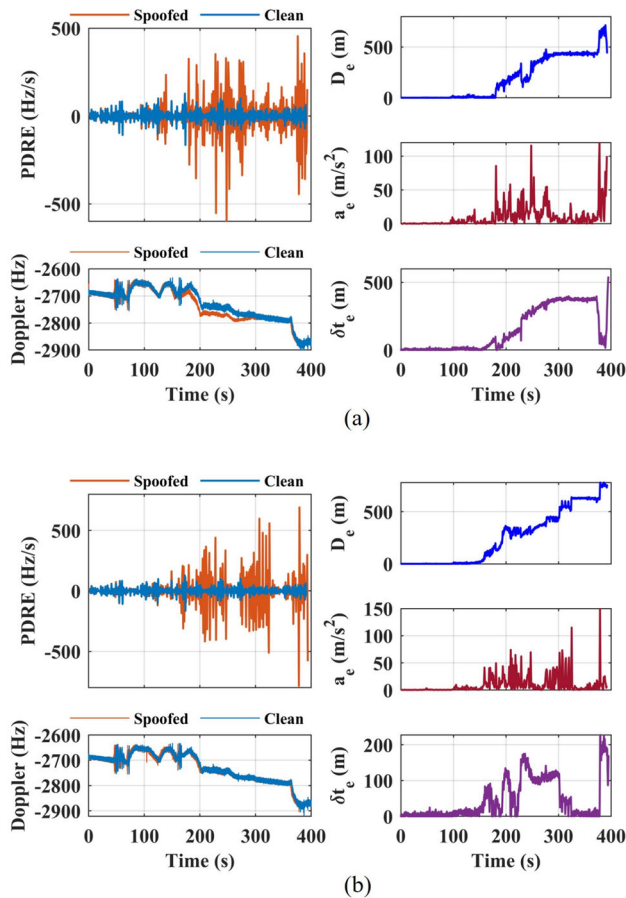
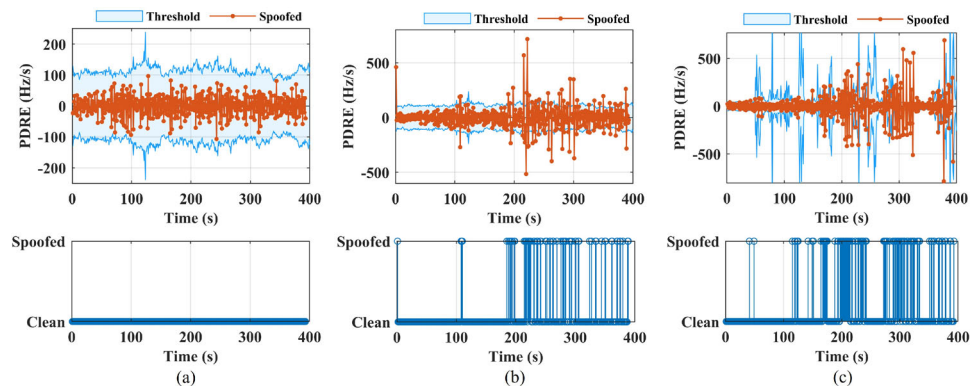


Fig. 10 PDRE and receiver-measured Doppler frequency obtained in the spoofing absence (clean) data and spoofing presence (spoofed) data, the magnitude of positioning error, acceleration error, and receiver clock error due to spoofing attack in PRN 15 of (a) TEXBAT ds5 (b) TEXBAT ds6

performance at the initial tracking stage. In TEXBAT ds6, the threshold undergoes significant fluctuations over time due to the dynamic motion of the receiver. Therefore, the proposed PDRE method demonstrates its capacity to distinguish a spoofing attack from noise interference caused by signal quality degradation due to receiver tracking errors,

Fig. 11 PDRE, moving threshold and corresponding detection output of (a) PRN 23 of TEXBAT ds2, (b) PRN 23 of TEXBAT ds4, and (c) PRN 15 of TEXBAT ds6



receiver dynamics, or environmental noise. However, two false alarms were observed around 50 seconds, likely caused by noise interference that led to an abnormal increase in the PDRE.

For commercial receivers lacking knowledge of receiver tracking loop parameters, such as noise bandwidth settings, these parameters can be roughly estimated by aligning the clean PDRE with the threshold, employing techniques such as maximum likelihood estimation [43]. Compared with the simulation tests, the PDRE in real-world experiments may experience abrupt fluctuations resulting from some unexpected interference, such as a random shift of signal power due to correlation or a large tracking error during the injection period. Note that these anomalies may enhance the PDRE method for spoofing detection.

6 Conclusion

The reliability and robustness of GNSS navigation are of great importance in public life and transportation systems. Spoofing attacks are among the most harmful interferences to the accuracy of GNSS receiver PVT solutions. The spoofing attack modes are updating over time. Therefore, a low-complexity spoofing detection method that is effective against both traditional overwhelming and sophisticated synchronous spoofing attacks is needed for mass-market receivers. This paper introduces the PDRE method for spoofing detection. The theoretical detection performance is analyzed in interference-free, multipath-affected, and spoofing-presence scenarios, proving effective in many typical spoofing attack cases. The detection algorithm is developed using the Neyman-Pearson decision rule, with the adaptive threshold set based on signal power. In addition, a spoofing attack simulator is proposed to estimate PDRE detection performance realistically and discuss the detection impact factors. Finally, the effectiveness of the PDRE method is validated using the public TEXBAT dataset. The experimental results demonstrate effective detection sensitiv-

ity when the receiver's positioning or timing begins to drift. In the future, the proposed PDRE's detection performance may be further validated through more field tests, focusing on its performance when the spoofer switches attack modes.

However, the proposed PDRE has limitations, as real-world results indicate that its detection performance can be significantly degraded in noisy environments. Therefore, countermeasures such as applying a lower tracking loop noise bandwidth or integrating external sensors could be implemented to enhance the stability of the receiver's navigation solution. It should be noted that the current threshold model does not explicitly include the uncertainty of the satellite ephemeris as an independent variance component. For applications when degraded broadcast ephemerides are used, the ephemeris-induced uncertainty should be modeled and incorporated into the threshold.

Appendix A Tracking loop noise

Assuming the utilization of a higher-order FLL for carrier frequency tracking, and the negligible impact of dynamic stress error, the Doppler frequency measurement noise is comprised of frequency thermal noise jitter. Accordingly, the estimation of the 1-sigma frequency jitter is [32]:

$$\sigma_{fFLL} = \frac{1}{2\pi T} \sqrt{\frac{4FB}{C/N_0} \left(1 + \frac{1}{TC/N_0}\right)} \quad (A1)$$

where F is 1 for high C/N_0 . B is the FLL noise bandwidth. For large early-to-late correlator spacing, the code-tracking jitter is [32]:

$$\sigma_{fDLL} = \sqrt{\frac{B_n}{2C/N_0} D \left(1 + \frac{2}{TC/N_0(2-D)}\right)} \quad (A2)$$

where B_n is the DLL noise bandwidth. D denotes early-to-late correlator spacing. The pseudorange measurement error is derived as:

$$\varepsilon_\rho = \sigma_{fDLL} \lambda_c \quad (A3)$$

λ_c is the code chip length.

Author Contributions All authors contributed to the study conception and design. Data collection and analysis and manuscript preparation were performed by Jingxiaotao Fang. Supervision, review and editing of the manuscript were performed by Bing Xu. All authors read and approved the final manuscript.

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Data Availability The University of Texas at Austin's radio navigation lab provides the TEXBAT datasets, which are available for download at the following link: <https://radionavlab.ae.utexas.edu/texbat>.

Declarations

Competing Interests The authors have no competing interests to declare that are relevant to the content of this article.

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